

Richard Pu

r3pu@uwaterloo.ca | linkedin.com/in/purichard | github.com/RichardPu | richardpu.ca

EDUCATION

University of Waterloo

Waterloo, ON

Candidate for B.A.Sc. in Computer Engineering (Honours, Co-op)

Sept. 2026 – June 2031

- **Stream 4 Co-op:** Seeking first 4-month internship (**Jan. 2027 – Apr. 2027**)

Thornhill Secondary School

Thornhill, ON

Ontario Secondary School Diploma

Sept. 2022 – June 2026

- 95% Cumulative Average (4.0 GPA) – Academic Honours Across All Disciplines

TECHNICAL PROJECTS

Smart Chess Board | NVIDIA Jetson Orin Nano, Python, C++, Flask

June 2026

- Migrated a physical chessboard from Arduino C++ firmware to an NVIDIA Jetson Orin Nano in Python, interfacing an OLED display over I2C/SPI, 64 addressable LEDs, and 10 tactile buttons via digital GPIO.
- Integrated Stockfish (offline AI) and the Lichess API (online play) with a real-time Flask dashboard for move history and custom LED themes, cutting start up time from 90 to 45 seconds.
- Redesigned and 3D-printed the enclosure for airflow, component access, and long-term repairability.

Real-Time CV “Red Light, Green Light” | Python, OpenCV, YOLOv8, TensorRT, Jetson

April 2026

- Built a self-officiating, multi-threaded Python game on a Jetson Orin Nano that tracks every player via YOLOv8-pose accelerated with TensorRT.
- Decoupled camera capture, pose inference, and game logic across separate threads, cutting inference latency from 300ms to 90ms and keeping display rendering responsive.
- Drove a servo-actuated head via PWM signals, integrating player clothing tracking and real-time audio TTS inside a Docker container for deployment.

BenumZombs – 2D Survival/Tower-Defense Game | Java, Swing/AWT

June 2025

- Architected a full Java/Swing game engine (loop, rendering, collision, resource, and building systems) supporting a day/night cycle and eleven upgradeable tower types.
- Designed an exponential zombie-multiplication mechanic where enemy count compounds over the course of a match, requiring the engine to handle steadily increasing entity loads without frame drops.
- Implemented collision detection using pairwise overlap checks between entities, and used polymorphic entity/tower classes to manage state transitions, resource costs, and automated targeting as entity count scaled up.

EXPERIENCE

Lifeguard & Swim Instructor

Vaughan, ON

City of Vaughan Aquatics

Sept. 2024 – June 2026

- Enforced pool safety regulations in a safety-critical environment, responding to first-aid incidents and de-escalating conflicts with the public.
- Delivered one-on-one and group swim lessons across age groups, completing 80+ hours of National Lifeguard/Swim Instructor certification (water rescue, First Aid/CPR-C).

LEADERSHIP & ACTIVITIES

Co-President, Tech Crew

Thornhill, ON

Thornhill Secondary School

Grades 9 – 12

- Led a student-run A/V team operating audio, lighting, and equipment for 30+ school events; grew active membership from 5 to 20 and built a shared calendar and inventory system that made the team self-sustaining.

Co-President, Announcements Team

Thornhill, ON

Thornhill Secondary School

Grades 9 – 12

- Identified a school-wide information gap and launched a digital Instagram announcements platform, reaching 50,000+ views in 90 days.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, HTML/CSS

Frameworks & Libraries: React, Flask, OpenCV, YOLOv8, TensorRT

Hardware & Tools: VS Code, NVIDIA Jetson Orin Nano, Arduino, Raspberry Pi, Docker, Git, GitHub, ESP32